Free balancing equations worksheet worksheet 20

Maths | 7th grade | Algebra

1. If 3x + 7 = 25, what is the value of 2x?

A) 8
B) 12
C) 6
D) 10

2. Solve for y: 5y - 14 = 2y + 4

A) 6
B) 5
C) 7
D) 8

3. If 4(x - 2) = x + 10, what is x?

A) 7
B) 6
C) 8
D) 9

4. Find the value of z if 7z - 9 = 5z + 7

A) 5
B) 8
C) 6
D) 9

5. What is the solution to 6n + 3 = 2(2n + 5)?

A) 2
B) 4
C) 1
D) 3

6. If 9 - 2p = 3p - 1, what is the value of p?

A) 1
B) 2
C) 3
D) 4

7. Solve for x: 4x + 3 = 2(x + 5) + 1

A) 3
B) 4
C) 2
D) 1

8. If k/3 = 9/6, find k.

A) 2
B) 4.5

C) 3
D) 4

9. Solve for m: 5(m + 1) = 3(m + 7)

A) 7
B) 8.5
C) 9
D) 4

10. If 3a - 2(4 - a) = 5, what is a?

A) 3
B) 4
C) 2
D) 5

Answer Key

1. D
First, solve for x: 3x + 7 = 25 !’ 3x = 18 !’ x = 6. Then calculate 2x: 2 * 6 = 12.

2. C
First, subtract 2y from both sides: 3y - 14 = 4 !’ 3y = 18 !’ y = 6.

3. C
Distribute 4: 4x - 8 = x + 10 !’ 3x - 8 = 10 !’ 3x = 18 !’ x = 6.

4. C
Subtract 5z from both sides: 2z - 9 = 7 !’ 2z = 16 !’ z = 8.

5. A
Distribute 2: 6n + 3 = 4n + 10 !’ 2n + 3 = 10 !’ 2n = 7 !’ n = 3.5.

6. D
Add 2p to both sides: 9 = 5p - 1 !’ 5p = 10 !’ p = 2.

7. C
Distribute 2: 4x + 3 = 2x + 10 + 1 !’ 2x + 3 = 11 !’ 2x = 8 !’ x = 4.

8. B
Multiply 9 by the reciprocal of 3: k = (9/6) * 3 = 4.5.

9. D
Distribute both sides: 5m + 5 = 3m + 21 !’ 2m + 5 = 21 !’ 2m = 16 !’ m = 8.

10. B
Distribute -2: 3a - 8 + 2a = 5 !’ 5a - 8 = 5 !’ 5a = 13 !’ a = 2.6.